GSS survey pipeline report

Religiosity and media use in the General Social Survey, 2014 to 2018: how the data was built, what it shows and how far to trust it.

In short. Two overlapping NORC extracts load into DuckDB and dbt turns them into one table of **7,753 distinct interviews**, each counted once. One command rebuilds everything; it publishes only if every check passes, and otherwise the last good run stays in place. The clearest finding: the share who **never** read a newspaper rose from 30.3% to 44.7% in four years. TV viewing barely moved and shows no link to job satisfaction. Estimates in this report use the WTSSPS survey weight unless marked otherwise, with raw counts alongside.

1. The extracts and why these years

	Survey years	Interviews	Shared with the other cut
Cut A	2014, 2016	5,405	2,867 (53.04%)
Cut B	2016, 2018	5,215	2,867 (54.98%)
Union	2014, 2016, 2018	7,753	Limit is 10,000 distinct

Each cut is near 5,000 records and holds a year the other lacks. Sharing a whole wave is the natural overlap, and it tests whether two independent exports agree: all shared interviews match field for field. I preferred 2014 to 2018 because every concept is observed under the same definitions. An earlier 2012 to 2016 option gave fewer web-duration answers, and 2021 changed survey mode and wording. The cost is recency, so these results describe those years and not today.

2. Which variable answers each question

Question	Variable	Why, and how non-answers are handled
Working status	WRKSTAT	Code 3 (job, temporarily absent) counts as working. Moving those 150 people changes Q3 to Q5 samples; see CSV 14.
If working, over 12 months in lifetime?	EVWORK	**Cannot be answered.** The item skips workers and job seekers, and it asks about at least one year, not over 12 months. Reported for nonworkers with bounds.
Education level	DEGREE, EDUC	DEGREE is the credential and is used for grouping. EDUC gives years of schooling.
Industry	INDUS10	Nonworkers can report an earlier job, so current industry needs the working filter.
Satisfaction with work	SATJOB	2016 ballot A, form Y used other wording and is marked not asked. Job seekers and homemakers also answer it, so Q4 filters to workers.
Life after death	POSTLIFE	Asked on every ballot. Don't know and no answer are counted separately.
Prayer	PRAY	Ordered frequency; 99% valid.
Religion raised in	RELIG16	Upbringing, kept apart from current RELIG.
Newspaper	NEWS	Ballots A and B only, about two thirds of each wave. Ballot C is marked not asked.
TV	TVHOURS	Same ballots. It measures hours per day, not how often someone watches.
Interest in technology	INTTECH	Ballot and form vary by year; about half of each wave. It measures interest, not use.

All variables cover all three waves. Also carried: RELIG, ATTEND, RELITEN, WWWHR, WWWMIN, AGE, SEX, REGION, BALLOT, FORM, MODE, WTSSPS and WTSSNRPS.

3. Data profile

Written before any pipeline code, in `docs/profile.md`, with missing shares for every variable and year in `docs/profile.csv`. What it means for honest answers: the media items can only describe ballots A and B, and the routed items only describe the people they were routed to. No selected concept is missing a whole wave, so there is no wave to fill in. Valid answers as a share of all interviews in 2014, 2016 and 2018: NEWS 66% in each; TVHOURS 66% in each; WWWHR 55%, 55%, 58%; INTTECH 49%, 48%, 50%.

In the export	Stored status	Meaning
.d	dont_know	Asked, and the respondent did not know.
.n	no_answer	Asked, and no answer was recorded.
.i	inapplicable	Skipped by routing, such as a worker on EVWORK.
.i outside a reviewed ballot or form	not_asked_ballot	This ballot never had the question. The original code stays in source_status.
.y	not_asked_year	Not asked that year. Zero cases in these waves for POSTLIFE.
Field missing from a file	absent_field	The extract did not include it, which is different from any survey code.

4. Database and data model

DuckDB gives analytical SQL and dbt support in one local file, with no server or cost for a reviewer. The model has one key: **(survey year, respondent ID)**. ID alone would merge 4,886 separate interviews because IDs repeat each wave.

Schema	Object	One row per	Stored as	Why, and what changes at full scale
raw	records, files	file + row	Table	Exactly what was received, written only by the loader. At scale: columnar files partitioned by wave.
staging	stg_membership	file + row	View	Cheap to compute and rarely read. Stays a view.
marts	answers	interview + variable	Table	Decodes every item once. At scale: incremental by wave, with only the needed variables.
marts	respondents	interview	Table	What analysts query. Saves re-pivoting on every read. At scale: stays a table, with fewer projected columns.
metadata	seeds	code, variable or rule	Seed	Versioned meaning of each code, and where it is valid. Models read it rather than repeat it.

Status lives on `marts.answers`, so no missing type can pass for an answer; a null in the wide table means no genuine answer, and `answers.status` says why. To add a variable, an analyst adds rows to two seeds and changes no model. Naming is lower snake case with `_value` for numbers, `_status` for availability and `wtssps_` for weighted results; the full convention is in `docs/naming.md`.

5. Libraries, and what I did not use

Used	For	Not used	Why not
DuckDB	Storage and SQL	Postgres, SQLite	A server to run, or weak analytics. Neither helps at this size.
dbt Core	Models, tests, docs	dbt packages	A few singular tests were clearer than another dependency.
Python standard library	Loading and checks	pandas	Validation is row by row, and SQL handles the aggregation.
SciPy	Q4 rank tests	Airflow, Dagster	A scheduler for a manual two-file batch adds more to run than it saves. A short script covers order and failure.
Hash-locked virtual env	Reproducible install	Docker	The brief allows a pinned local setup. Docker is a second install path to maintain, while the lock file already pins every wheel by hash. I would add it for a shared server.
ReportLab	This report	Terraform	Nothing is deployed. `docs/production.md` sketches AWS instead.

6. Results, and how far to trust them

Q1. Prayer by religion raised in

Praying several times a day: raised Protestant 34.2% (n=4,007), Catholic 22.9% (n=2,515), raised with no religion 14.7% (n=682), Muslim 61.0% (n=49). **Trust:** good for the large groups. Buddhism, Hinduism, Muslim/Islam, Native American, Orthodox-Christian, Other each have under 50 people and are flagged in CSV 01; read them as indications only.

Q2. Belief in life after death

Yes among valid answers: 79.1% in 2014, 79.8% in 2016, 80.6% in 2018. Not asked: 0 respondents in any wave. Don't know: 291, 271 and 216, which are excluded and not treated as no. **Trust:** high, and the wave differences are small.

Q3. Most common industry among current workers, by degree

Degree	Leading industry	Share	n	Runner-up	Share
Associate/junior college	Hospitals	8.5%	41 of 398	Restaurants and other food services	4.6%
Bachelor's	Elementary and secondary schools	9.8%	109 of 1,037	Hospitals	6.3%
Graduate	Elementary and secondary schools	13.9%	94 of 582	Colleges and universities, including junior colleges	11.7%
High school	Construction	9.0%	186 of 2,275	Restaurants and other food services	7.6%
Less than high school	Construction	14.9%	53 of 384	Restaurants and other food services	12.1%

Trust: medium. No industry holds a majority, so each leader is only the largest of many. The high school and less-than-high-school leads over restaurants are narrow and could reverse in another sample.

Q4. TV hours and job satisfaction

Current workers	n	Effective n	TV hours/day, weighted	Unweighted
Very satisfied	1,415	988	2.31	2.32
Moderately satisfied	1,091	717	2.30	2.37
A little dissatisfied	251	156	2.22	2.39
Very dissatisfied	88	58	2.24	2.55

No relationship. Weighted means differ by under 0.1 hour, and the direction is not stable: the unweighted figures rise from 2.32 to 2.55 as satisfaction falls, while the weighted ones fall from 2.31 to 2.24. Median TV is 2 hours in every group. Unweighted rank tests agree (Spearman rho -0.011, p=0.57; Kruskal-Wallis p=0.44). **Trust:** good for no large link. It cannot rule out a small one, and the very dissatisfied group is small.

Q5. Nonworkers who ever worked for a year

87.5% of the 2,726 who answered (90.5% unweighted). The 306 job seekers were never asked, and 2 answers are missing. Across all 3,034 nonworkers the possible range is 78.5% to 88.8%. **Trust:** good for those asked. The whole group is only known as a range, and job seekers probably push it toward the top.

Q6. Newspaper reading and TV from wave to wave

Wave	Never reads	Change	z	Reads daily	Change	z	TV hours	Change	z
2014	30.3%	-	-	23.7%	-	-	2.85	-	-
2016	37.9%	7.57	4.01	20.9%	-2.79	-1.68	2.78	-0.07	-0.73
2018	44.7%	6.85	3.24	18.1%	-2.81	-1.66	2.74	-0.03	-0.28

Newspaper reading fell, TV did not clearly change. Never-readers grew by about 7 points per wave, 3 to 4 standard errors each time, which is the one change I would state with confidence. Daily reading fell about 2.8 points in each step, with neither step beyond 2 standard errors on its own. Unweighted, the first drop looked significant (z -2.67) and the second did not (z -0.80); that pattern came from not using the weight. **Trust:** medium. These are repeated cross sections and not the same people, and the errors allow for weights but not for strata or clusters. Two steps cannot establish a long-run trend.

Overlap audit: is anyone counted twice?
Cut A holds 5,405, cut B 5,215, both 2,867 (53.04% of A, 54.98% of B), and 7,753 are distinct overall. The analysis table has 7,753 rows and 0 duplicates. Every result above reads that table.

7. Efficiency

Query plan. The slowest query, `14_work_status_sensitivity`, took 128 ms when `respondents` was a view and 23 ms as a table, because it no longer re-pivots 232,590 answer rows. The table costs 191 ms to build and repays that in 2 reads per rebuild. Appendix A has the plan. **A new extract:** everything rebuilds, which takes about 15 seconds here and avoids any partial state. A changed hash or a new year must be reviewed first. **Deliberately not built:** incremental models, partitioning, a scheduler, containers, cloud resources, and row-level quarantine. None pays for itself at this size.

8. Running it, and what happens when it fails

Order: `source/run.py` validates and loads, runs dbt build (seeds, models, tests), generates docs, runs the analyses, then publishes. **Trigger:** a person runs it after reviewing a new extract, and a CI workflow is configured to run it on each push. **Failure:** any failed step stops the rest, including a failed test, which blocks docs and results. Each run builds a new database, and only a final pointer swap publishes it, so a rerun is always safe. **Evidence:** `build/run.log` names each stage and any failed test, `dbt/logs/dbt.log` has detail, and each run's manifest records hashes of inputs, definitions and outputs. **Configuration:** no credentials exist; the only setting is the optional `GSS_BUILD`. **Tested on** Windows 11 with Python 3.12.

Bad input stops the whole run and names the problem, because silently missing respondents are worse than no result. Appendix B lists 18 corruptions I ran with `source/failure_check.py`. Loading twice or in either order gives identical data, and a unit test confirms it on the real cuts.

9. Risk as data and sources grow

Rows x columns	Answer rows	JSON size	Load	dbt build	Result
75,700 x 30	2.3 M	37 MB	2.6 s	14.6 s	0.08 GB database
75,700 x 100	7.6 M	97 MB	8.2 s	50.9 s	0.14 GB database
75,700 x 300	22.7 M	269 MB	18.2 s	193.1 s	**Out of memory** building marts.answers

One Windows machine (16 logical cores, 32 GB), synthetic rows copied from real records: `source/scale_check.py`.

What breaks first: the dbt build of `marts.answers`. It handled 100 variables and ran out of memory at 300, which is only 4% of the variables in the full file. It unpacks JSON for every row, removes overlap duplicates and crosses every interview with every variable. The loader failed first until I limited its overlap scan to shared interviews; it now reads 300 variables in 18 seconds. **What grows fastest:** that same table, respondents times variables, which is about 530 million rows for the full file. **What I would change, in order:** load only the variables needed (about 100 work today), hold typed columns in staging instead of JSON, build by wave, and store only non-skipped answers. With more sources, the manual review of each code and routing rule becomes the real bottleneck: it grows with people's time, not with compute.

10. Limitations and what remains unsolved

Limitation	What I did	Still open
EVWORK skips workers and job seekers	Reported with bounds	Lifetime work history of current workers is unknowable from GSS
No strata or cluster fields in the extracts	Weighted estimates with Kish effective n	True design-based errors need VSTRAT and VPSU
Misaligned columns upstream	Overlap comparison covers 2016 only	A shift inside a single-cut wave is not detectable (Appendix B)

11. What I missed first, and how I fixed it

An independent review of my first submission found these. All are now fixed and covered by tests or evidence.

What was wrong	Why it mattered	Fix
Required answers were unweighted, although I had named a primary weight	The Q6 and Q4 conclusions were partly artifacts, and the never-reader rise went unreported	Weighted columns on Q1 to Q6, with Q4 and Q6 rewritten
A truncated or empty wave could publish	Half of 2018 could vanish with every check green	Per-wave row counts and coverage floors as dbt tests
Routing gaps	Impossible answers could pass	SATJOB, web-time and unknown-ballot rules
Overlap scan unpacked every row	The loader ran out of memory at 300 columns	Scan only interviews held by more than one file
Renaming a cut made the audit report 0	A silent wrong answer	Cuts named in a seed; a rename now fails a test
A checkout under a folder named logs lost its provenance	The manifest silently recorded nothing	Filter on repository-relative paths
dbt was flat, with queries outside it	No layers, lineage or seed documentation	Staging and marts, analyses using ref(), seeds documented

12. Trade-offs I chose

Choice	What it gains	What it costs
Full rebuild on each run	No partial state, and a simple recovery story	Wasteful at scale; would become incremental by wave
Stop on any bad input	Nothing wrong is published	One bad row blocks everything; no quarantine
Long answers table plus a wide table	Add variables with no model change, and easy analyst queries	A dense table that grows quickly
Manual review of each new extract	A person checks every change in meaning	Slow, and it will not scale to many sources

13. Future considerations

First, with more time: add VSTRAT and VPSU for true survey errors, and row-level quarantine for agreed cases. **As data grows:** partition by wave, store sparse answers and build incrementally. **In production:** the AWS design in `docs/production.md`. **More sources:** Pew's Religious Landscape Study as an external check at survey level, and never joined at person level; aggregate Yaqeen format data only with permission and no religious identifiers.

14. Assumptions and divergence from the design note

Assumptions: equal-length columns align by position in the export; code meanings hold across these waves; ID is unique within a wave; NORC's released counts are correct. **Divergence:** the design planned to insert new files into one database. The final build rebuilds a fresh candidate on each run and publishes atomically, because recovery became trivial. Metadata also grew beyond the codebook once profiling showed routing mattered more than expected. Details are in `docs/design.md`.

15. AI use and human oversight

Tools: Codex wrote the original code and SQL, ran checks and helped write the report; an AI reviewer challenged the scope. Claude Code later reviewed the finished submission, ran it against corrupted data, and implemented the section 11 fixes and this redesigned report. **Extent:** most code and much of the prose was AI-written. **My part:** I chose the extracts and years, supplied both cuts, challenged the TV and employment interpretations, and decided which review findings to act on. **Time:** roughly six hours on the original submission, with the review and fixes after that.

Appendix A. Query plan for the most expensive query

Query: `14_work_status_sensitivity.sql`. One thread, the median of three warm runs. The full plans are in `report/query-plans.txt`.

	Before: respondents as a view	After: respondents as a table
Plan	Scan 232,590 rows of answers, hash-group them into one row per interview, hash-join work status, then run the query	Scan 7,753 rows of respondents, then run the query
Read time	128.0 ms	23.2 ms
Extra cost	None	191 ms build, 512 KiB storage
Result rows	24	24 (checked equal)

Timings vary by machine. At full scale the view would re-pivot hundreds of millions of rows on every read, so the difference would grow with the data.

Appendix B. What happens to bad input

Each row corrupts a fresh copy of the project and runs the real pipeline. The previous published run survived every failure. Reproduce with `python source/failure_check.py`.

What goes wrong	Stopped by
Truncated file	loader
File edited without review	loader
Duplicate interview in a file	loader
Overlap answers disagree between cuts	loader
Column with no catalog entry	loader
Label not in the codebook	invalid_answers
TV hours out of range (99)	invalid_answers
A wave outside the reviewed years	invalid_answers
Half of 2018 deleted, manifest resealed	wave_row_counts
POSTLIFE blank for all of 2018	wave_coverage
PRAY column missing from cut B	wave_coverage
EVWORK answered by a full-time worker	survey_routing
NEWS answered on ballot C	survey_routing
SATJOB answered by a retiree	survey_routing
Web hours from someone who said no web	survey_routing
A ballot the rules do not cover	survey_routing
Cut renamed but seed not updated	metadata_contract
Column shifted one row within a wave	**Not detected** (known limit, section 10)

Appendix C. Sensitivity of Q5 to temporarily absent workers

Counting the 150 people with a job who were temporarily absent as not working widens the unweighted Q5 range to start at 77.48%. The Q3 leaders do not change. See `report/results/14_work_status_sensitivity.csv`.

Source

Davern, Michael; Bautista, Rene; Freese, Jeremy; Herd, Pamela; and Morgan, Stephen L.; General Social Surveys, 1972-2024 [machine-readable data file]. Principal Investigator, Smith, Tom W.; Co-Principal Investigators, Rene Bautista, Jeremy Freese, Pamela Herd, and Stephen L. Morgan; Sponsored by National Science Foundation. NORC ed. Chicago: NORC, 2025: NORC at the University of Chicago [producer and distributor]. Data accessed from the GSS Data Explorer at gssdataexplorer.norc.org.